

Biochemistry lab

Experiment (7)



EXPERIMENT 7

PREPARATION OF AN ENZYME EXTRACTION

- ✚ **Purpose:** to determine the optimum conditions for polyphenol oxidase (PPO) enzyme, and how to inhibit it using different inhibitors.
- **polyphenol oxidase (PPO):** also known as tyrosinase, an enzyme (catalyst) found in humans and plants, used to convert tyrosine to melanine pigment through an oxidation reaction.

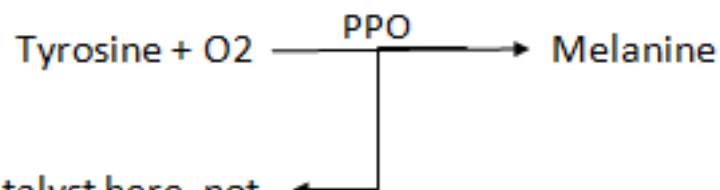
In humans:

- PPO used to convert tyrosine to melanine pigment, this pigment is responsible for the skin color (brown color)
- humans lacking polyphenol oxidase (tyrosinase) exhibit albinism (بياض لون الجلد)

In plants:

- when we cut a fruit (apple, banana, potato...), and expose it to air (O_2), a brownish color will develop. This color is produced by melanine pigment (from a reaction catalyzed by PPO)
- the brownish color is evidence for oxidation reaction which leads to food corruption (فساد الأغذية).

The reaction in general:

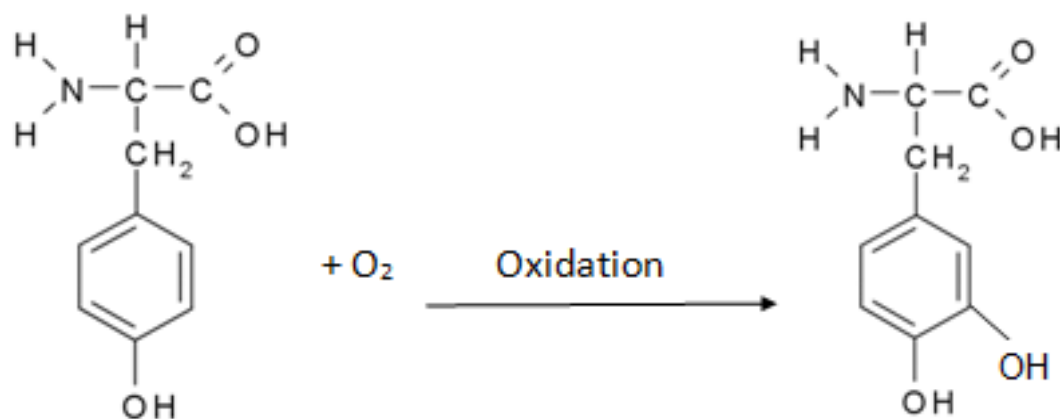


PPO is a catalyst here, not

A reactant (without it the reaction won't happen.)

Note: all PPO reactions are oxidation reactions

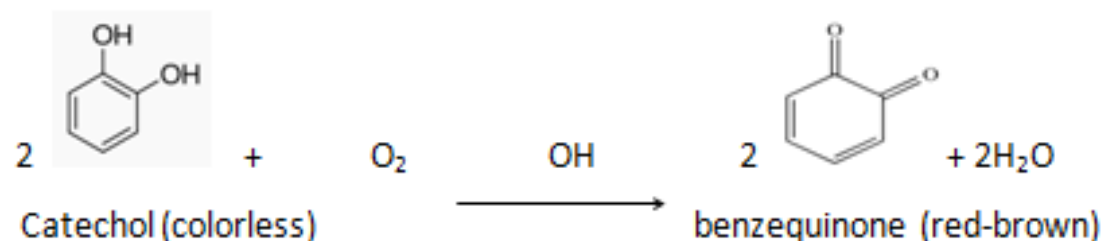
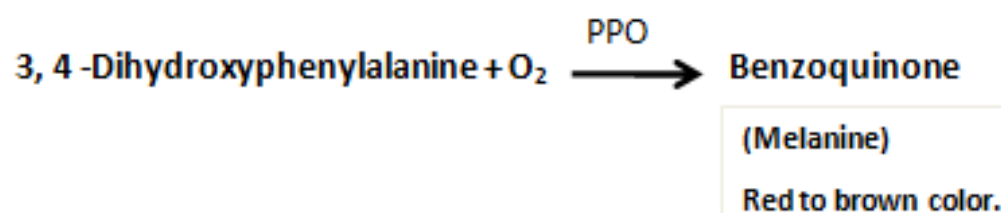
In details:



(Tyrosine)

3, 4 -dihydroxyphenylalanine

(Contains 2OH)



- This enzyme contains a copper ion (Cu⁺²) in its active site (essential for its work) and this enzyme can be inhibited (deactivated) by **two ways**:
- 1) Remove the copper from the enzyme using phenylthiourea (which binds to copper ion.)
 - 2) Remove O₂ from the medium of reaction using ascorbic acid (**Vitamin C**).

"يعني، يمكن تثبيط عمل انزيم ال PPO بطريقتين، الطريقة الأولى هي إضافة Phenylthiourea مما يزيل Cu المهم جدا لعمل الإنزيم. الطريقة الثانية هي إضافة فيتامين C الذي يعتبر مضاد للأكسدة، يمنع الأكسدة عن طريق إزالة الأوكسجين من الوسط."

-we take PPO from potato. PPO is one of thousands of enzymes in potato, so, to inhibit a class of enzymes (Phosphatases) we add fluoride in the form of NaF to the potato preparation.

"المقصود أن البطاطا غنية بالإنزيمات، لذا سنحاول أن نتبط عمل جميع الإنزيمات إلا الانزيم المطلوب (PPO)، عن طريق إضافة الفلور على شكل NaF عند تحضير البطاطا."

In our experiment, we will study the effect of different factors on PPO enzyme and its work.

How to prepare the enzyme extract (مستخلص الإنزيم التي استخدمناه)?

1. set up a funnel lined with six layers of cheese cloth in a funnel rack placed over a 100 ml beaker immersed in an ice bath

Cheese cloth: قطعة قماش مميزة

2. peel a small or one-half of a large potato
 3. Dice potato and add to homogenizer, add 50 ml of NaF solution, note that NaF is a poison (PPO is soluble in NaF, NaF prevents oxidation.)
 4. Blend it for 1-2 min
 5. Pour homogenate through cheesecloth – lined funnel and collect soluble lysate in a beaker
- ** In this experiment you will need 20 ml of lysate ****
6. Wrap waste in paper towels and throw away
 7. Keep lysate on ice to minimize proteolysis of enzyme and the production of colored products within lysate.

*** In our experiment, we will use only the supernatant of the beaker***

"ملحوظة للتوضيح: الخطوات السابقة كانت لاستخلاص PPO من البطاطا، لم نقوم بفعل أي من الخطوات السابقة داخل المختبر، كان الإنزيم جاهزاً."

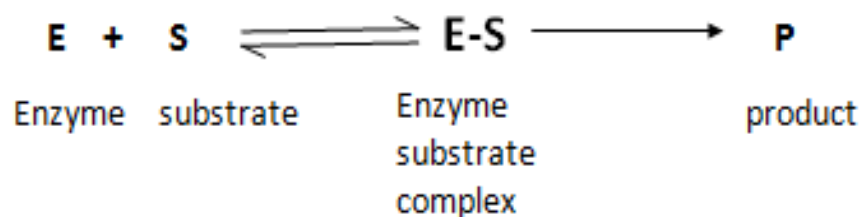
Let's start with **the factors affecting the enzyme**

1- kind of substrate
(substrate specificity)

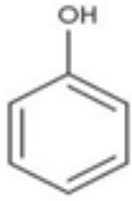
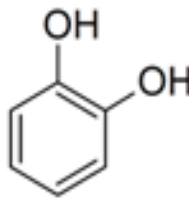
- Different substrates differ in:

1) Number of OH 2) Position of OH.

- General reaction:

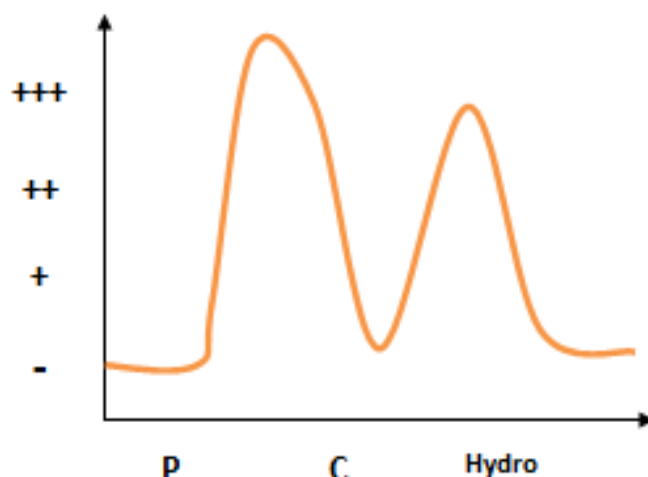


The Experiment:

		
15 drops phenol	15 drops Catechol	15 drops Hydroquinone
		
Monohydroxyphenol (Negative -)	Ortho- dihydroxyphenol (+++) at 37°	Para dihydroxyphenol (+)

Few drops of enzyme extract

1. label 3 test tubes
2. add few drops of enzyme extract
3. Tube A: 15 drops of phenol (0.01M)
Tube B: 15 drops of Catechol (0.01M)
Tube C: 15 drops of Hydroquinone (0.01 M)
4. shake tubes and place in water bath (37°)



As # of OH increase, color becomes darker

As OHs is closer, color becomes darker.

Ortho > Meta > Para.

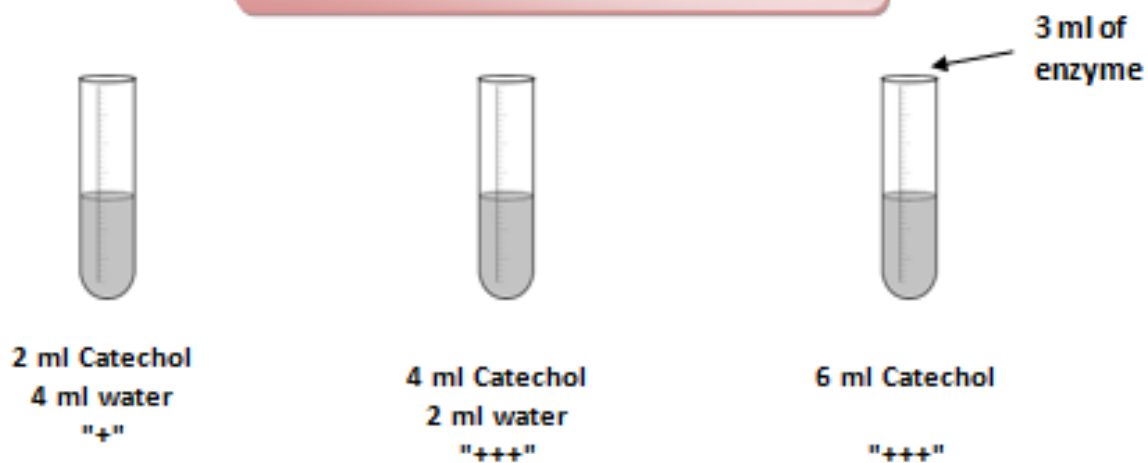
هذا استنتاجنا عاملين, العامل الأول هو عدد
OH, Monophenol أعطى (-) رغم
وجود OH, لأن واحدة فقط لا تكفي
لإظهار اللون.

العامل الثاني هو قربهما عن بعضهما,
عندما يكونان Ortho يكون اللون أقوى
منه عندما يكونان Para.

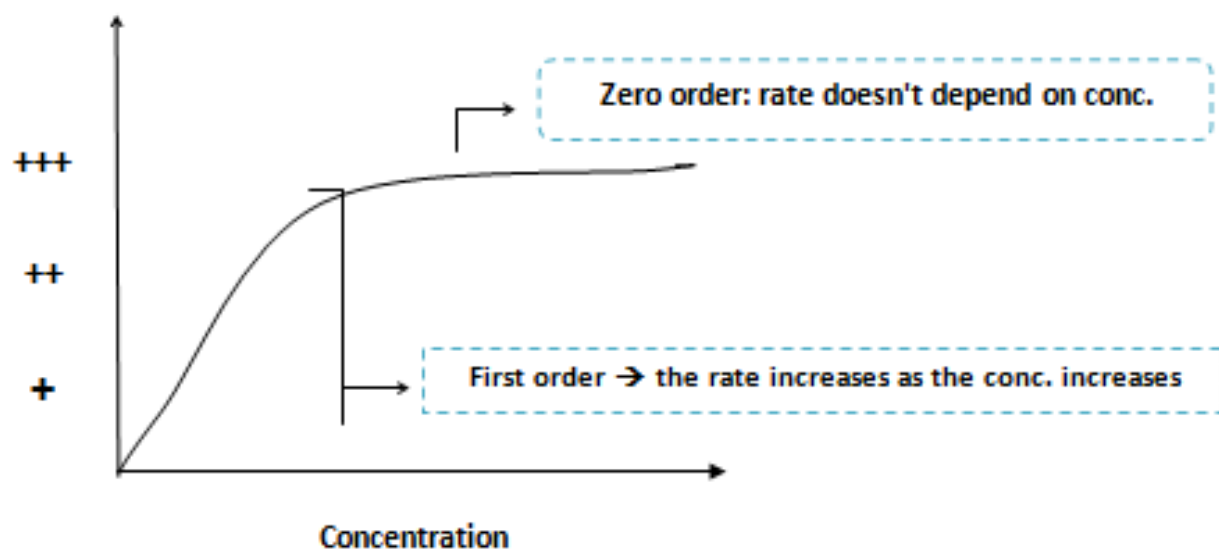
-Oxidation reaction occurs for
Di and Tri hydroxyphenol

-meta form of diphenol may
give a mediate color (++)

2-Concentration of substrate:



At 37 C for 20 (5 min interval)



As concentration increases, color becomes darker

- until a limit, at that limit color stops becoming darker. Because we have limited number of enzymes

"تماما مثل إشارة المرور التي تسمح بمرور سيارتين فقط، مهما كان الازدحام فإن الإشارة ستسمح بمرور سيارتين في الثانية، أي أن السرعة لا تعتمد على تركيز السيارات."

→ **STEPS:**

1. label 3 test tubes (we use 0.01 Ml catechol)
2. A: 2 Ml Catechol + 4 Ml water
B: 4 Ml Catechol + 2 Ml water
C: 6 Ml Catechol
3. Add 3 Ml of enzyme to each test tube.
4. Place in water bath (37°C), Shake every 2-3 min (or plump, to gain O₂)
5. Examine tubes at 5 minutes interval for 20 minutes and record the result.

3- Effect of temperature on enzyme action

First step: 15 drops of enzyme

Second step: 15 drops of Catechol

0°C



(-)

Inactivation
temporary

37°C



The optimum
temperature
(+++)

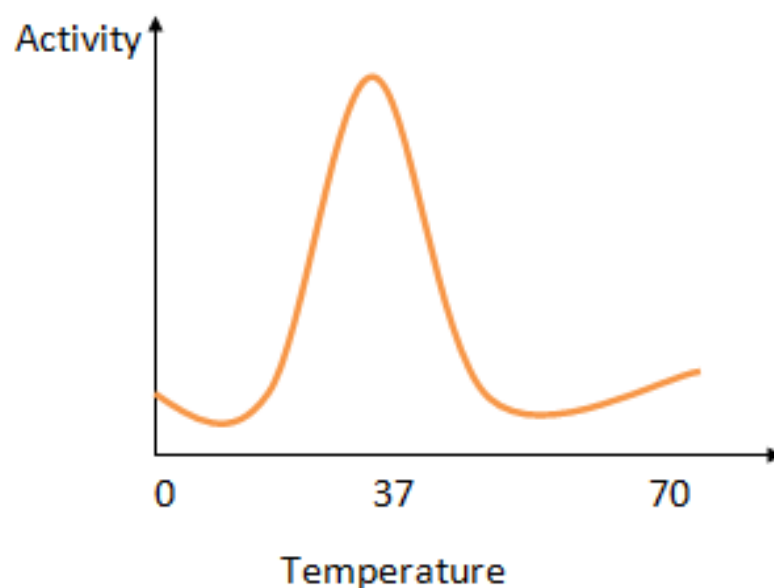
70°C



Denaturation
(effect of heat)

Irreversible
denaturation (-)

Not temporary

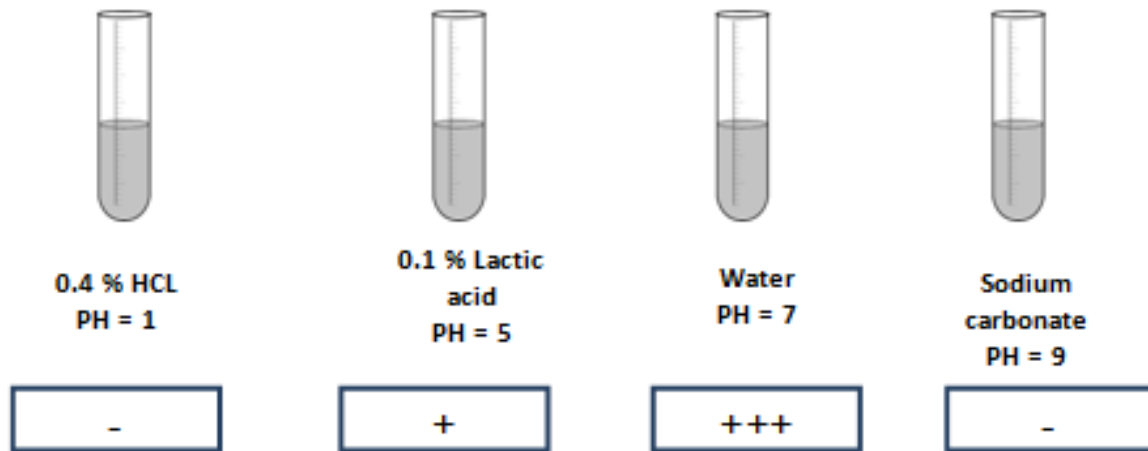


→ STEPS:

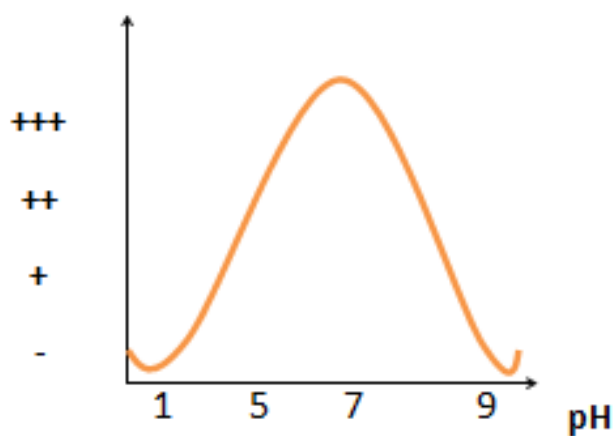
1. Label 3 test tubes
 2. Add 15 drops of enzyme extract on each test tube.
 3. Place one test tube in water bath at 70°C, one in the fridge at 0°C, one at normal 37°C
 4. Add 15 drops of 0.01 Ml catechol solution to each test tube
 5. Quickly remove the tube, shake gently and return immediately to its proper temperature
 6. Wait 15 min, examine for color change, without removing the tube from its temperature, and record the results.
- remember that adding Catechol to the enzyme before changing is temperature will give us positive test though. The reaction will happen, so we change PPO's temperature alone, not the mixture's.

"الحرارة تؤثر على عمل الانزيم, اذا كان الانزيم عند درجة منخفضة فإنه لن يعمل, واذا كان عند درجة عالية فإنه لن يعمل أيضا وسيفقد البروتين خواصه هنا,"

4 -Effect of PH



- Optimum pH = 7
- Enzyme works at pH = (5-7)



عندما نريد تجربة عامل معين، دائماً نقوم بتثبيت العوامل الأخرى على الحالة optimum. هنا مثلاً أريدنا تجربة درجة الحموضة، إذا سئمت درجة الحرارة على 37، وثبتت نوع المادة على Catechol، وثبتت التركيز، وهكذا.

steps:

1. Prepare 4 test tubes as following:
A: 2ml of 0.4% hydrochloric acid (HCl). pH = 1
B: 2ml of 0.1% lactic acid solution. pH = 5
C: 2ml of distilled water. pH = 7
D: 2ml of 0.5% sodium carbonate. pH = 9
2. Add 15 drops of 0.01M catechol to each tube.
3. Add 15 drops of enzyme extract to each tube.
4. Shake well, place in water bath (37°C). (Or plump)
5. Wait 15 min, examine the color and record the results.

5) Effect of inhibitor:

- the activity of PPO can be inhibited by the presence of sodium azide (NaN₃), thiourea, cinnamic acid, L-phenylalanine and L-tryptophan.

Without inhibitor (+++)



At temperature 37°C

With inhibitor (-)



Steps:

1. Add 15 drops of 0.01M sodium azide or thiourea to a test tube containing 15 drops of enzyme extract.
2. In another tube, add 15 drops of enzyme extract.
3. Add 15 drops of 0.01M catechol and put the tubes in water bath (37c).
4. Compare the colors and record the results.

Visual activity of polyphenol oxidase:

- (-): no color change
- (+): light (faint) color
- (++): definite color
- (+++): dark, deep color

وضع فقائيع

● Note that we can increase the concentration of O^2 by plumping in test tube.

Q: in preparation of enzyme extract, what is the solvent that we added to PPO?

Answer: NaF (sodium fluoride).

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THE END



Questions

Q1) which of the following will give us the test results ?

- A) Phenol
- B) Catechol
- C) Hydroquinone

Q2) Arrange the following according to the darkness of the color produced?

- A) Para, Ortho, Meta.
- B) Meta, Ortho, Para.
- C) Ortho, Meta, Para

Q3) which of the following will give us the test result?

- A) 3mL of Catechol+ 1mL of H₂O
- B) 3mL of Phenol
- C) 3mL of Catechol only
- D) 4mL of Phenol

Q4) what is the optimum temperature and pH for PPO?

- A) 0, 10
- B) 37, 7
- C) 36, 6

Q5) which of the following statements are correct about the effect of pH on PPO ?

- A) We will get a +ve result on pH= 7 only.
- B) At pH= 2 the sample will be denatured.
- C) At pH= 12 the sample will be denatured.
- D) Two answers are correct.

1-B 2-C 3-C 4-B 5-D

